



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. (General) Degree

First Year Semester II Examination – March 2013

CHE 1203 – ORGANIC CHEMISTRY I

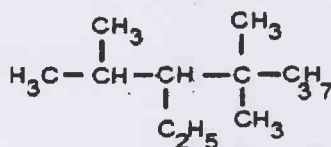
Answer any four questions.

Time: 02 hours

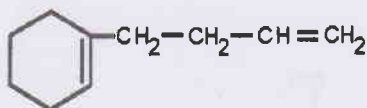
01.

(a). Give IUPAC names of the following compounds

i).



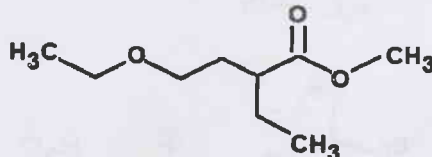
ii).



iii).



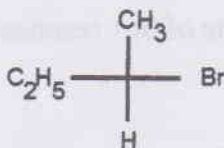
iv).



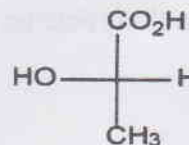
(10 marks)

(b). Give IUPAC names of the following compounds with R/S, *cis/trans* or E/Z configurations

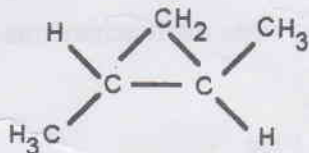
i).



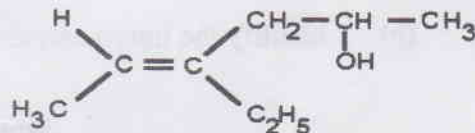
ii).



iii).



iv).



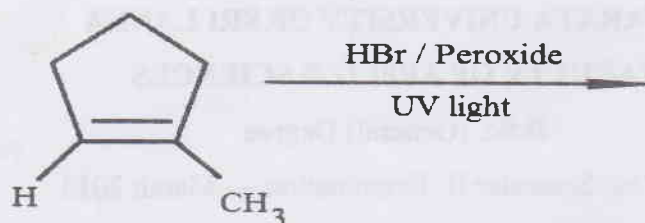
(12 marks)

(c). Draw the structure of the compound (E,Z)-2,4-heptadienal.

(03 marks)

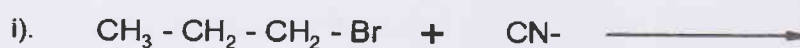
02.

(a). Identify the product and write the mechanism of the following reaction



(04 marks)

(b). Identify the products of the following Nucleophilic Substitution reactions and briefly discuss the reactions

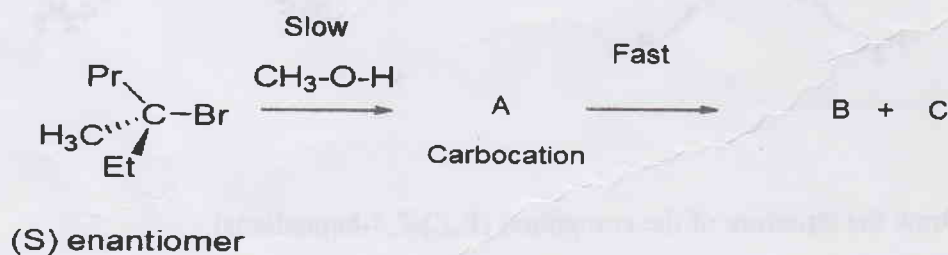


(09 marks)

(c). Write a short note on the factors influencing the rate of S_N1 reaction

(07 marks)

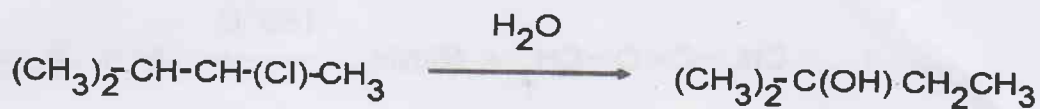
(b). Identify the intermediates & final products and write the mechanisms of the reactions.



(05 marks)

03.

- (a). Write the complete mechanism of the following reaction



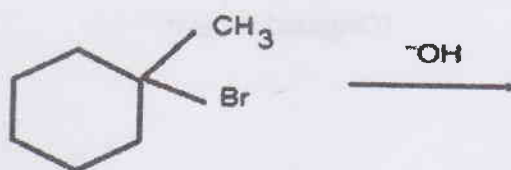
(06 marks)

- (b). Methyl chloride reacts with OH^- produces Methanol. Write the mechanism of this nucleophilic substitution reaction and draw the relevant energy diagram.

(08 marks)

- (c). Discuss the products of the following elimination reactions

i).



1-Bromo-1-methylcyclohexane

ii).



1-Chloro-1,2 diphenylethane

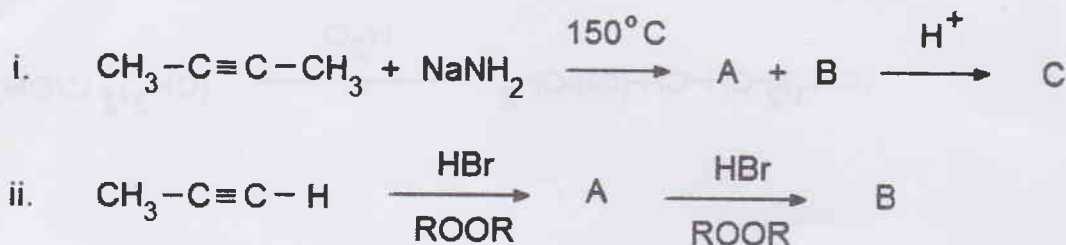
(08 marks)

- (d). When compound (A) treated with $\text{O}_3 / \text{H}_2\text{O}$, equal moles of acetic acid and 3-methyl butanoic acid formed. Identify (A) and write the reaction.

(03 marks)

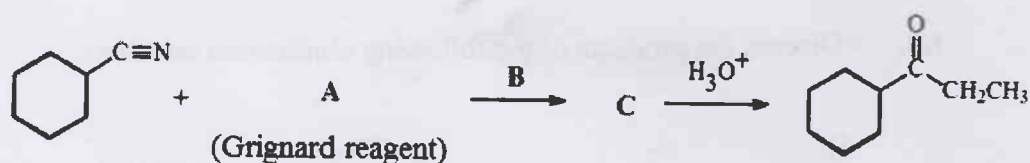
04.

(a). Identify A, B and C and complete the following reactions

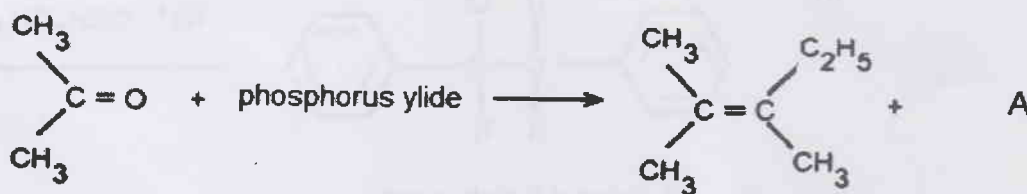


(08 marks)

(b). Identify A, B and C and complete the following reaction



(05 marks)

(c). The **Wittig reaction** can be given as follows

Identify the structures of the phosphorus ylide and the by-product (A)

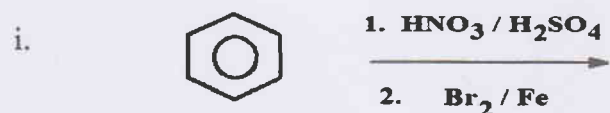
(05 marks)

(d). Propyl benzene can be obtained when compound (A) follows **Wolff-Kishner reduction**. Further, when (A) treated with HCN followed by HCl/H₂O, 2-hydroxy-2-methyl-3-phenylpropanoic acid can be obtained. Write all reactions and identify (A).

(07 marks)

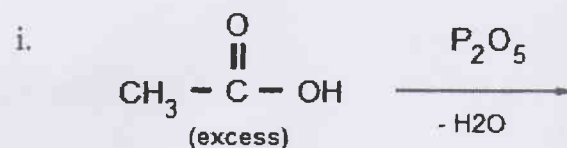
05.

(a). Compare the following two reactions



(04 marks)

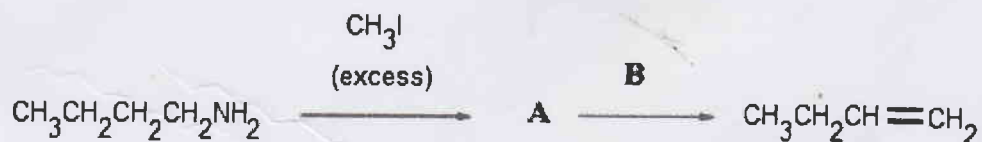
(b). Complete the following reactions



(05 marks)

(c). Discuss the synthesis of Ethyl amine ($\text{CH}_3\text{CH}_2\text{NH}_2$) using **Gabriel reaction**

(08 marks)

(b). The **Hofmann elimination reaction** of butylamine can be expressed as follows.

- Identify intermediate product (A) and reagents / condition (B).
- What is the purpose of (A) in the reaction
- Propose a mechanism for the reaction.

(08 mark)

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